

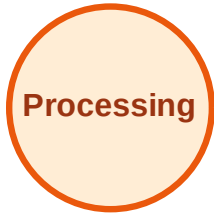
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



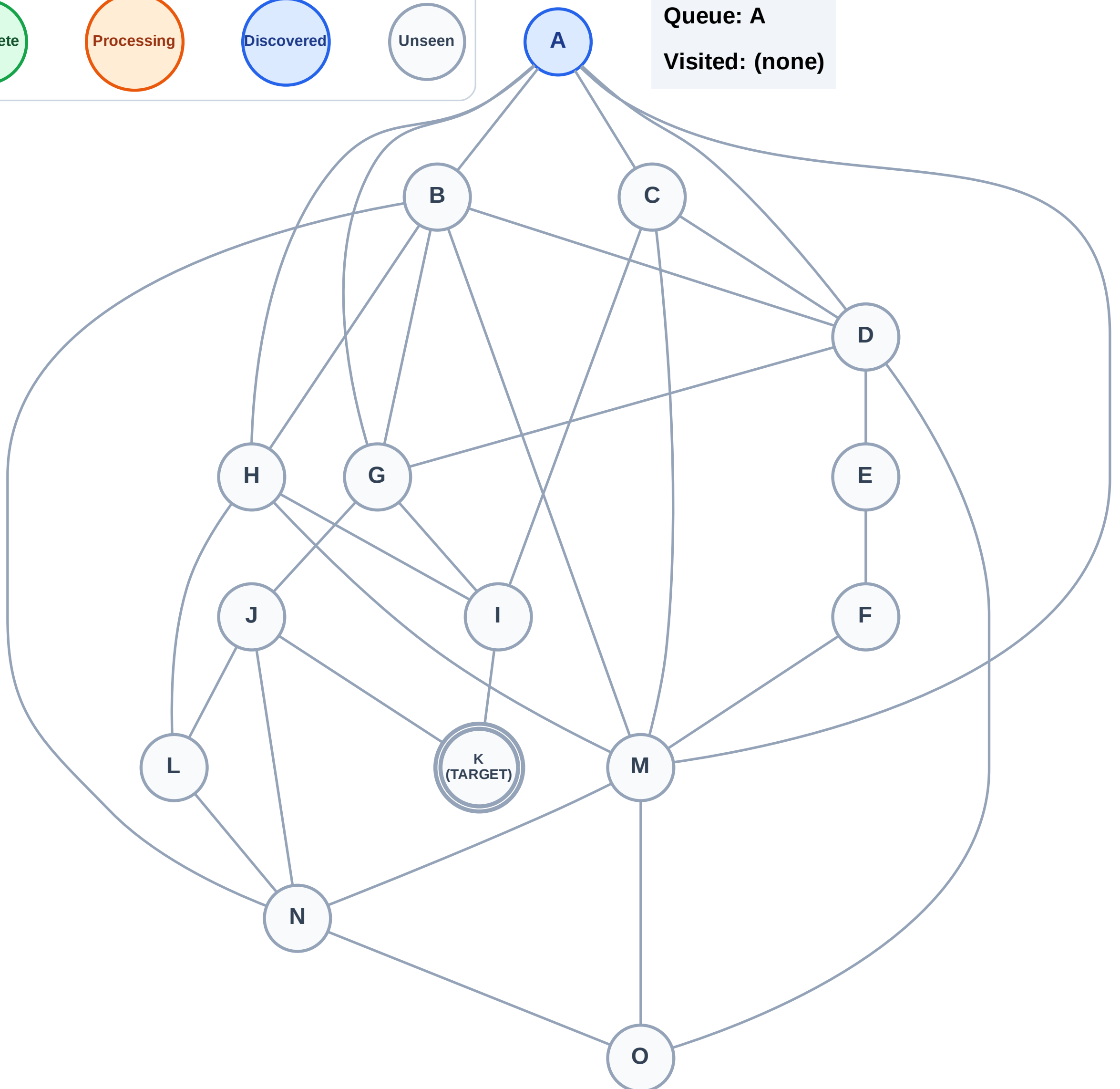
Discovered



Unseen

Queue: A

Visited: (none)



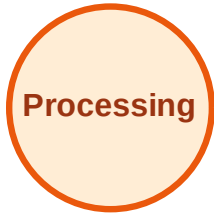
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



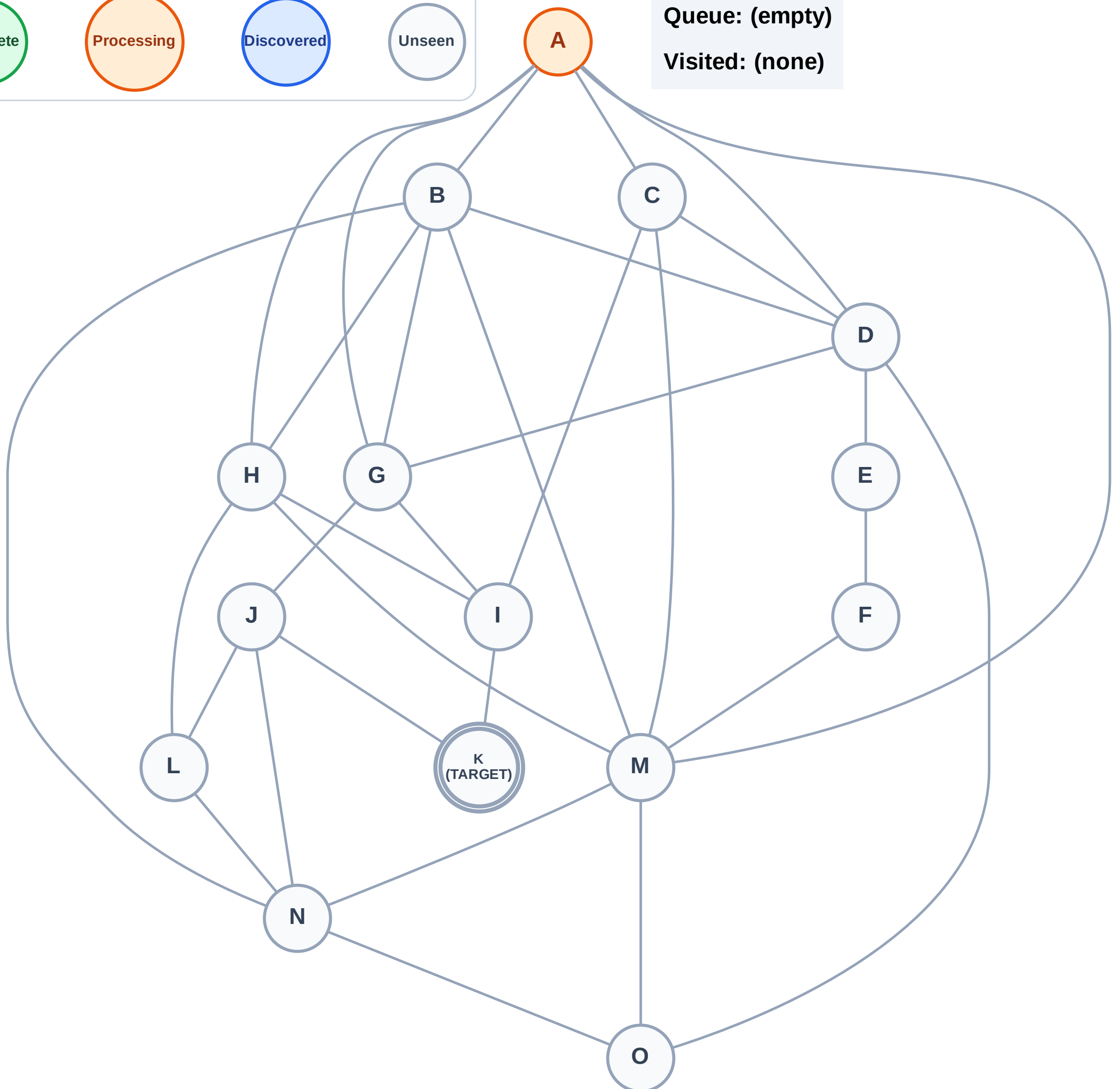
Discovered



Unseen

Queue: (empty)

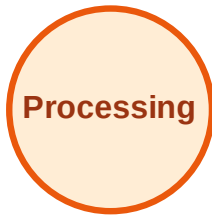
Visited: (none)



BFS Traversal

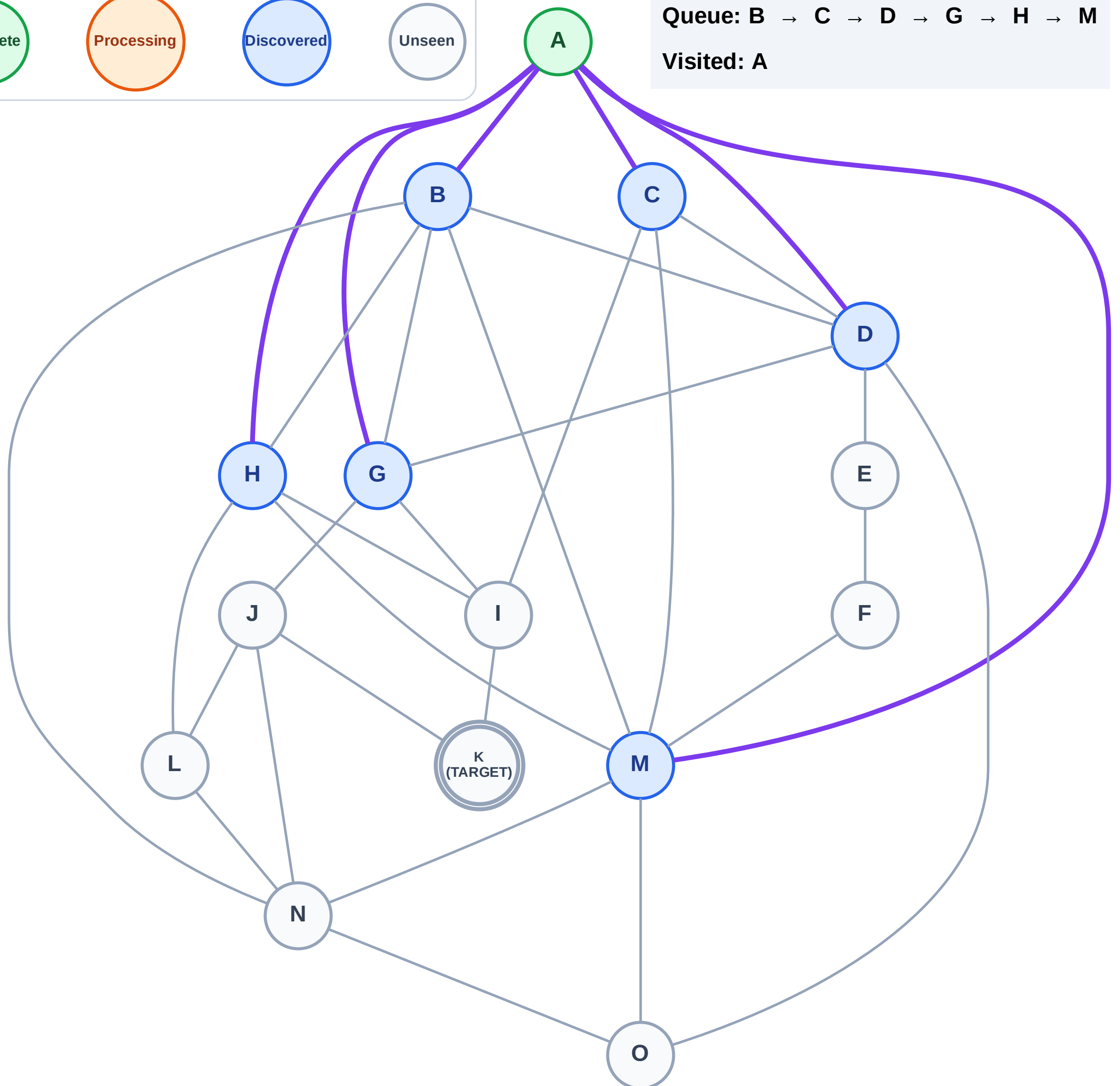
Step 3: Discover B, C, D, G, H, M from A

Legend



Queue: B → C → D → G → H → M

Visited: A



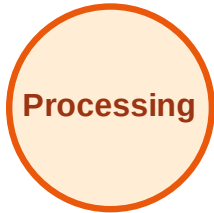
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



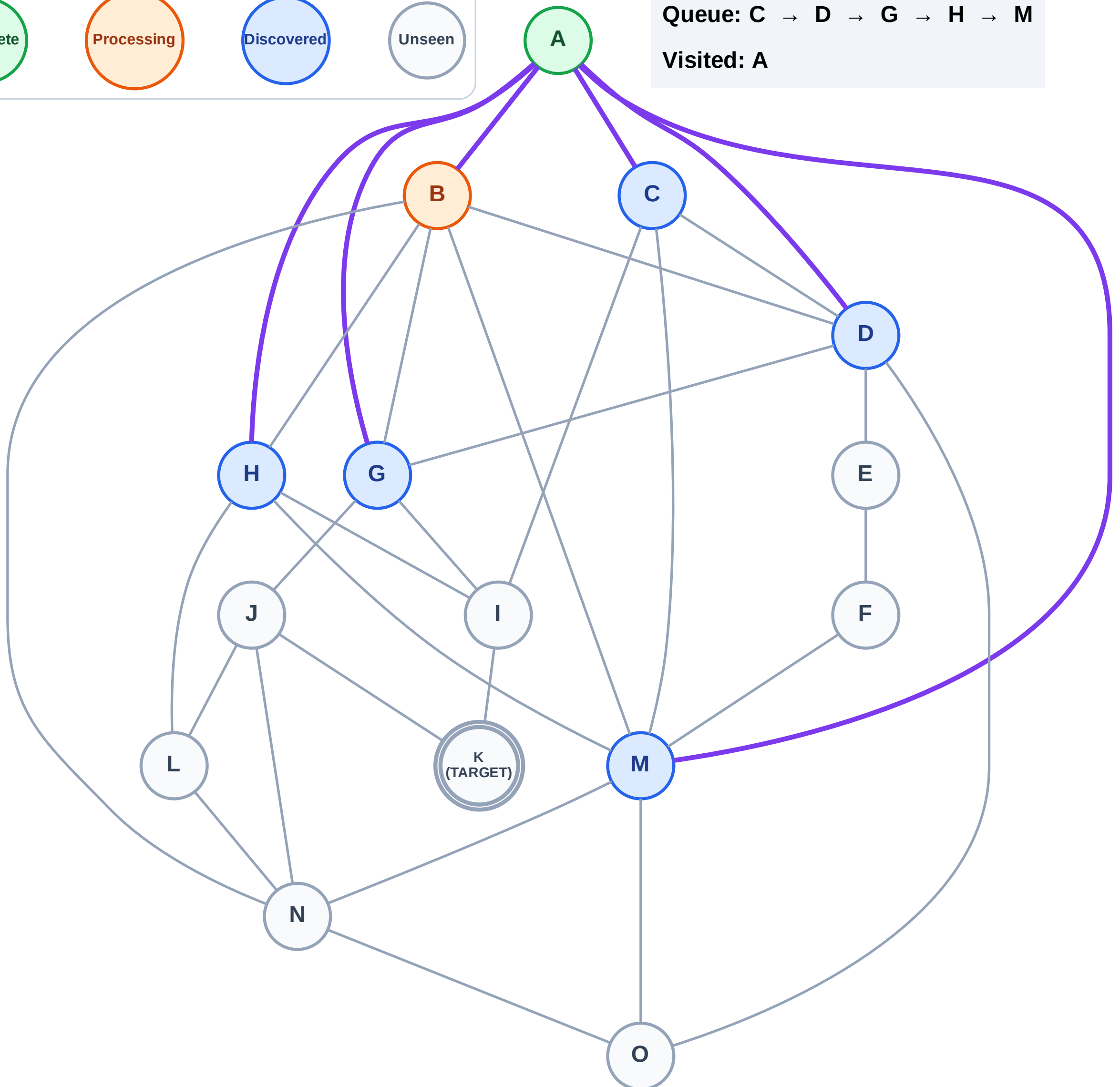
Discovered



Unseen

Queue: C → D → G → H → M

Visited: A



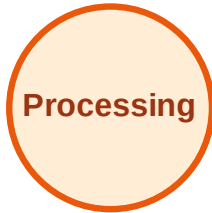
BFS Traversal

Step 5: Discover N from B

Legend



Complete



Processing



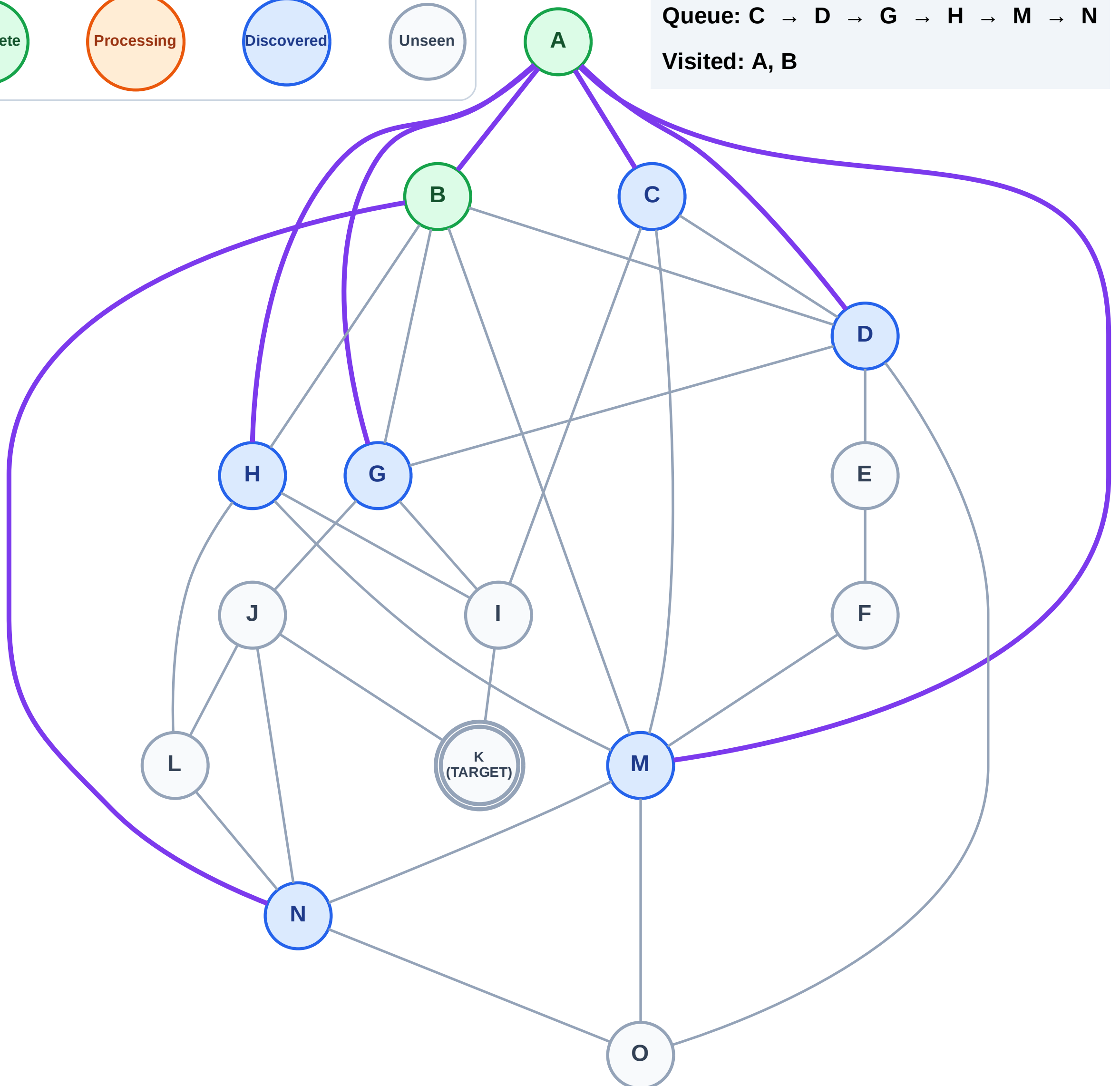
Discovered



Unseen

Queue: C → D → G → H → M → N

Visited: A, B



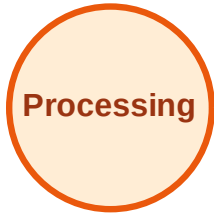
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



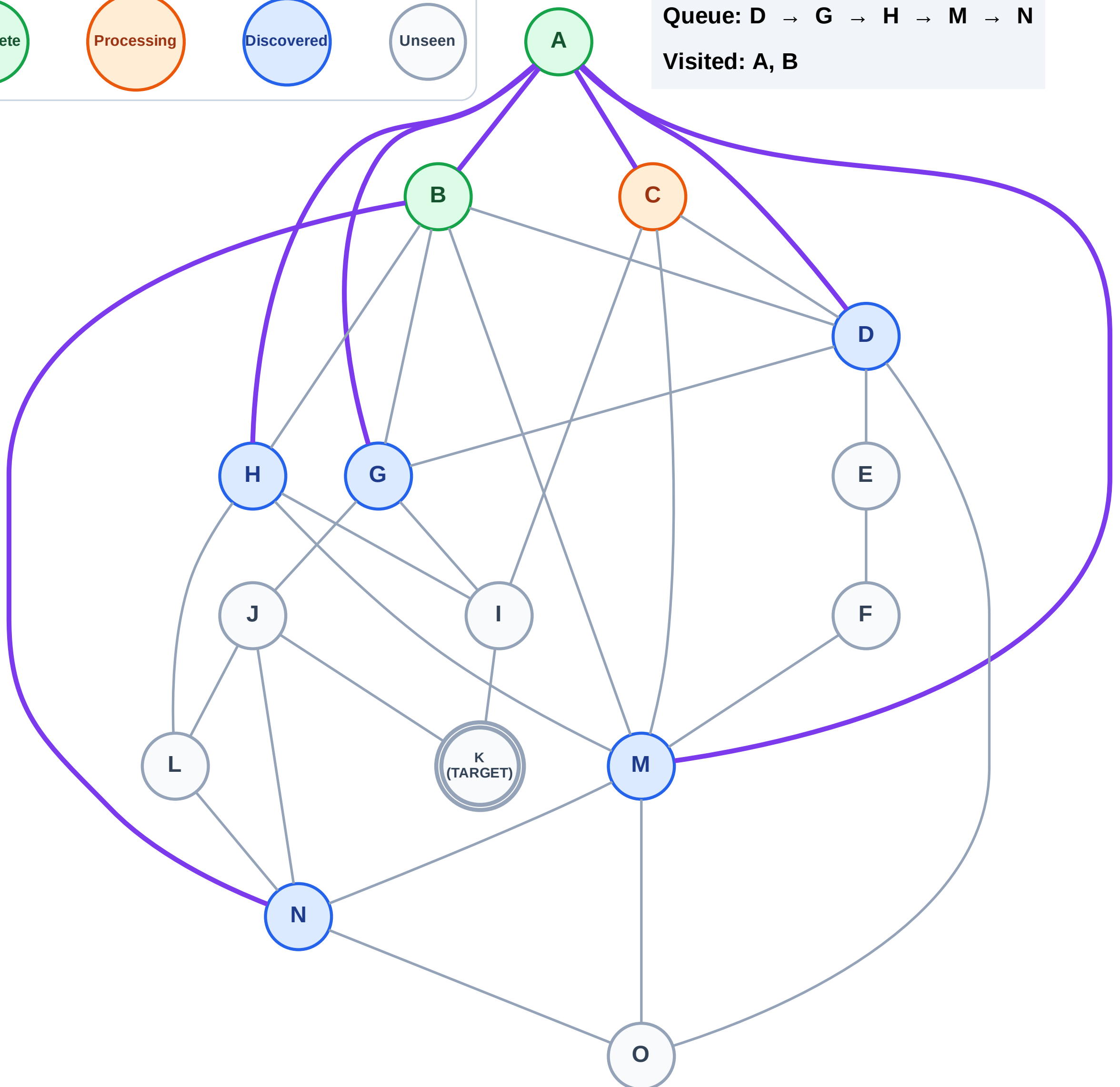
Discovered



Unseen

Queue: D → G → H → M → N

Visited: A, B



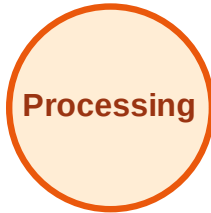
BFS Traversal

Step 7: Discover I from C

Legend



Complete



Processing



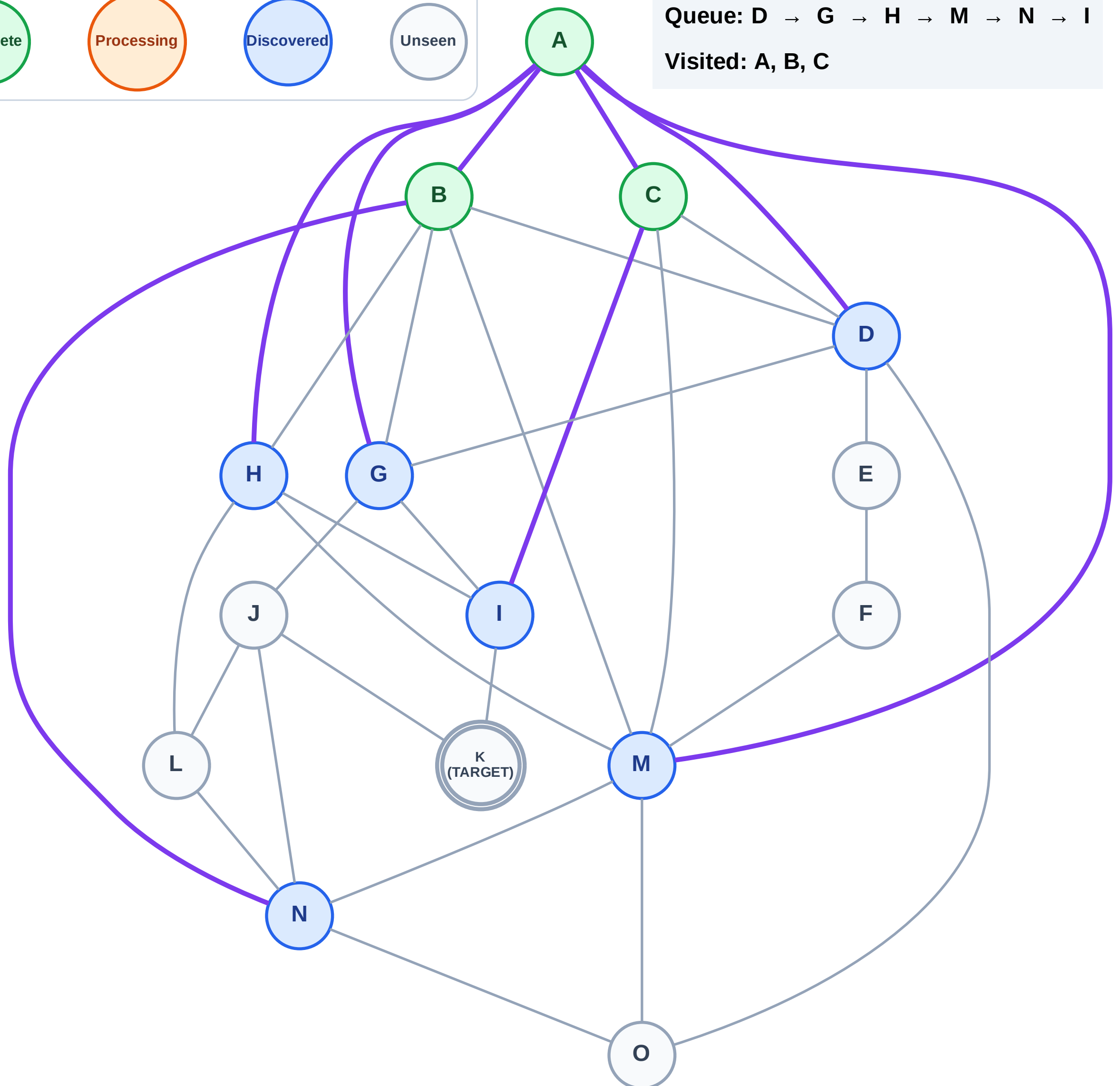
Discovered



Unseen

Queue: D → G → H → M → N → I

Visited: A, B, C



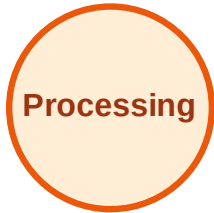
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



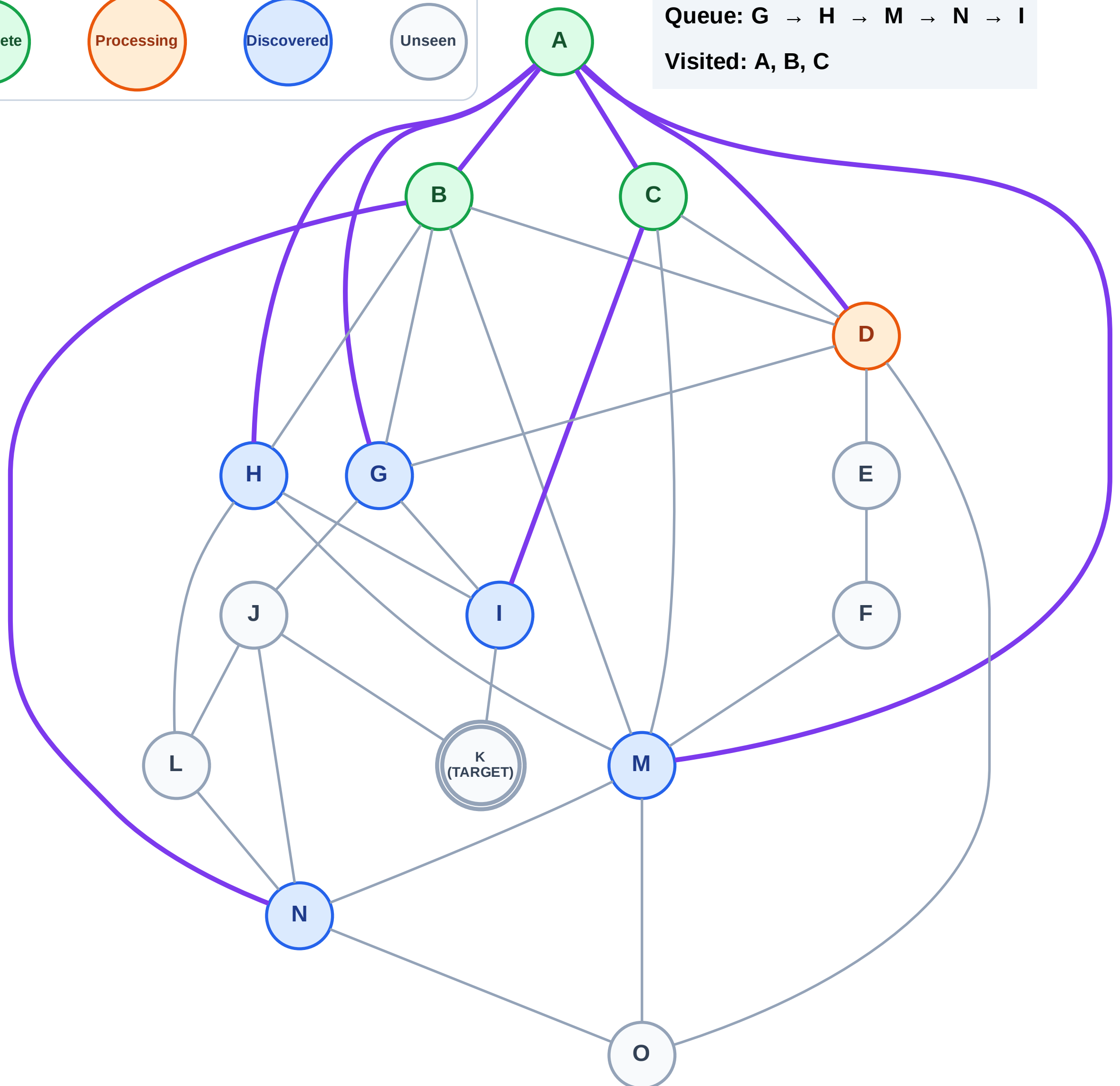
Discovered



Unseen

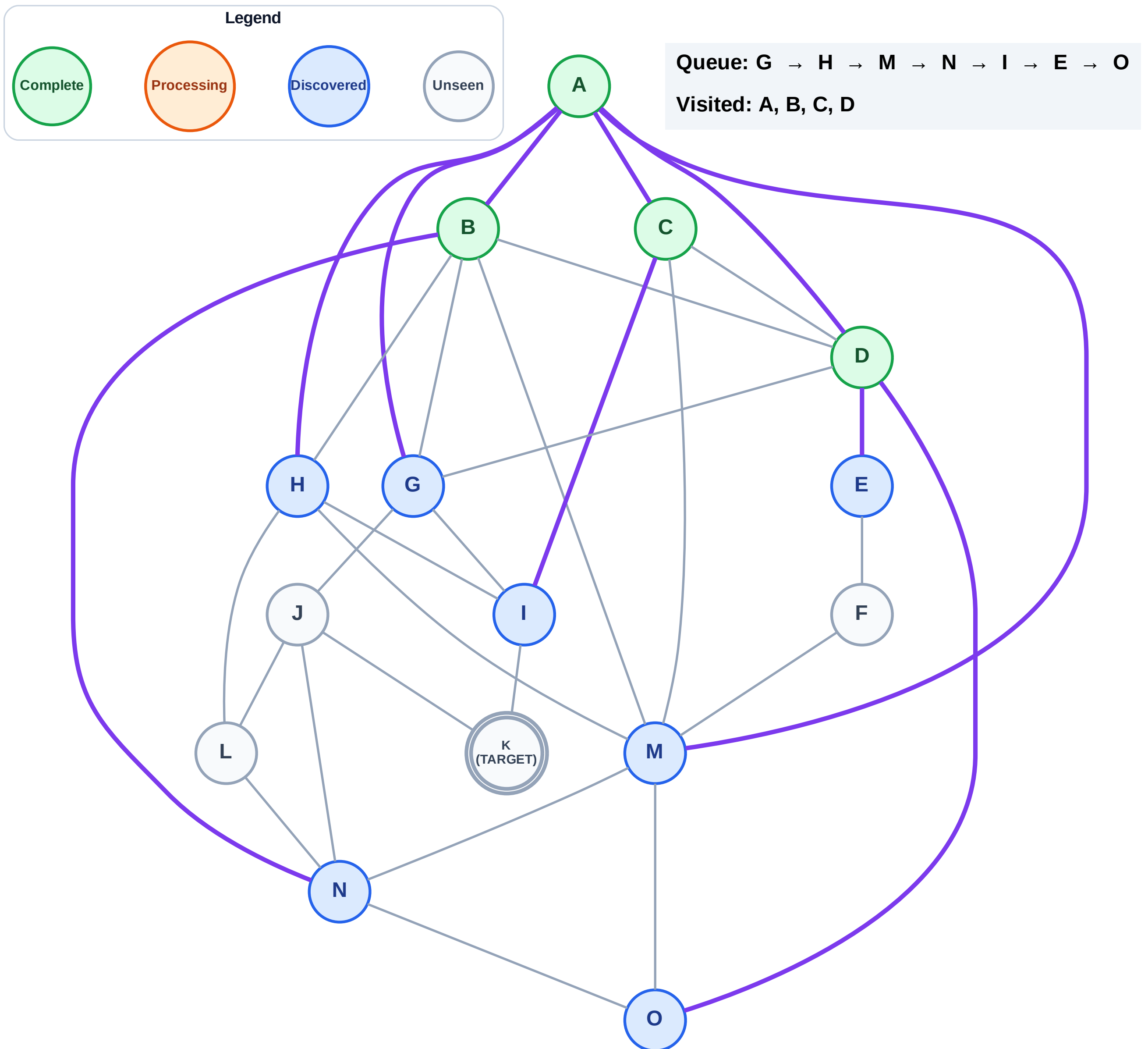
Queue: G → H → M → N → I

Visited: A, B, C



Step 9: Discover E, O from D

Step 9: Discover E, O from D



BFS Traversal

Step 10: Process node G

Legend

Complete

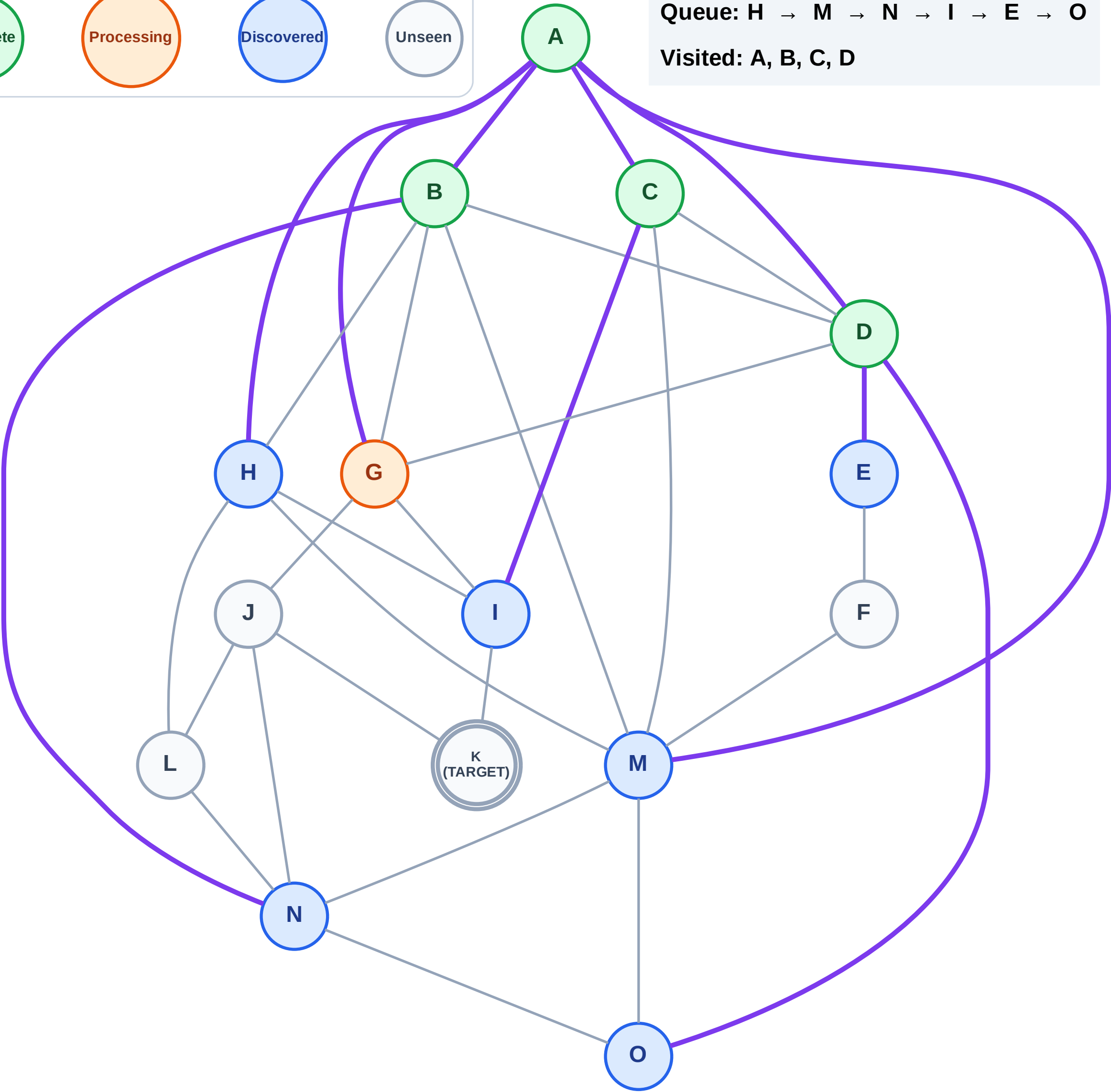
Processing

Discovered

Unseen

Queue: H → M → N → I → E → O

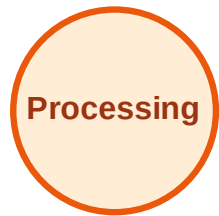
Visited: A, B, C, D



BFS Traversal

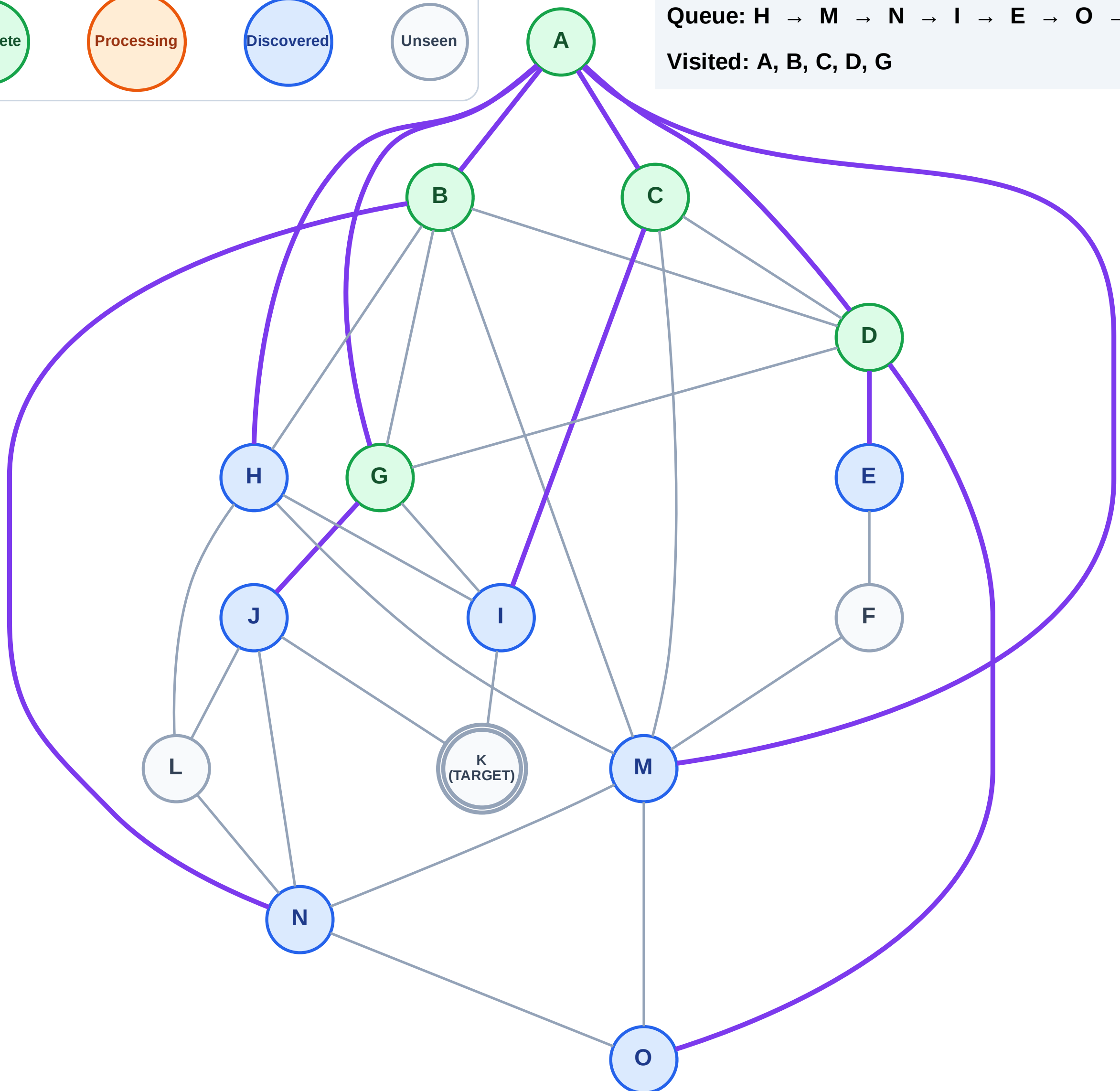
Step 11: Discover J from G

Legend



Queue: H → M → N → I → E → O → J

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node H

Legend

Complete

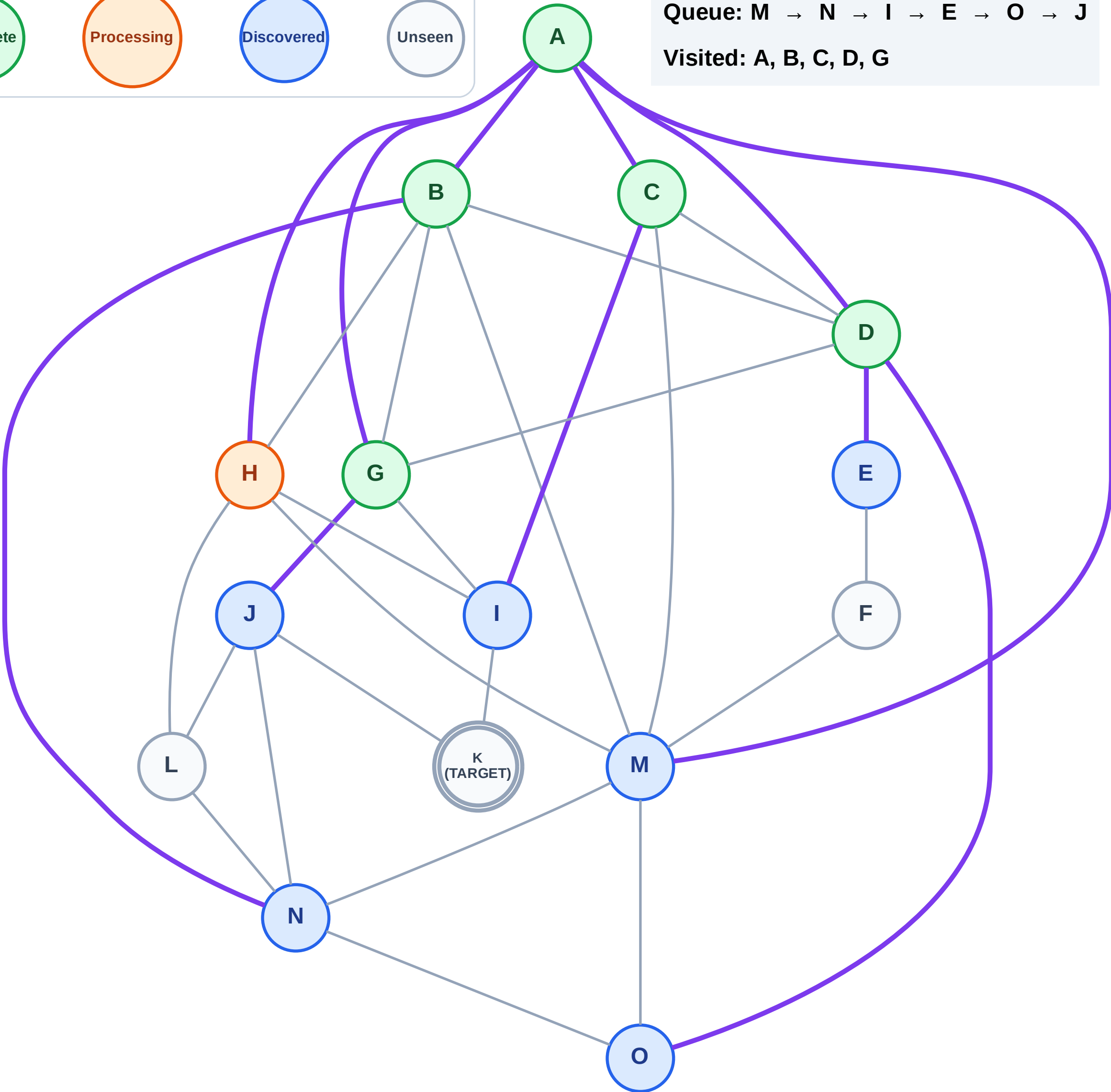
Processing

Discovered

Unseen

Queue: M → N → I → E → O → J

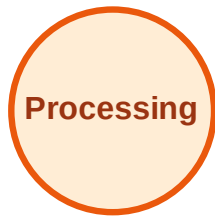
Visited: A, B, C, D, G



BFS Traversal

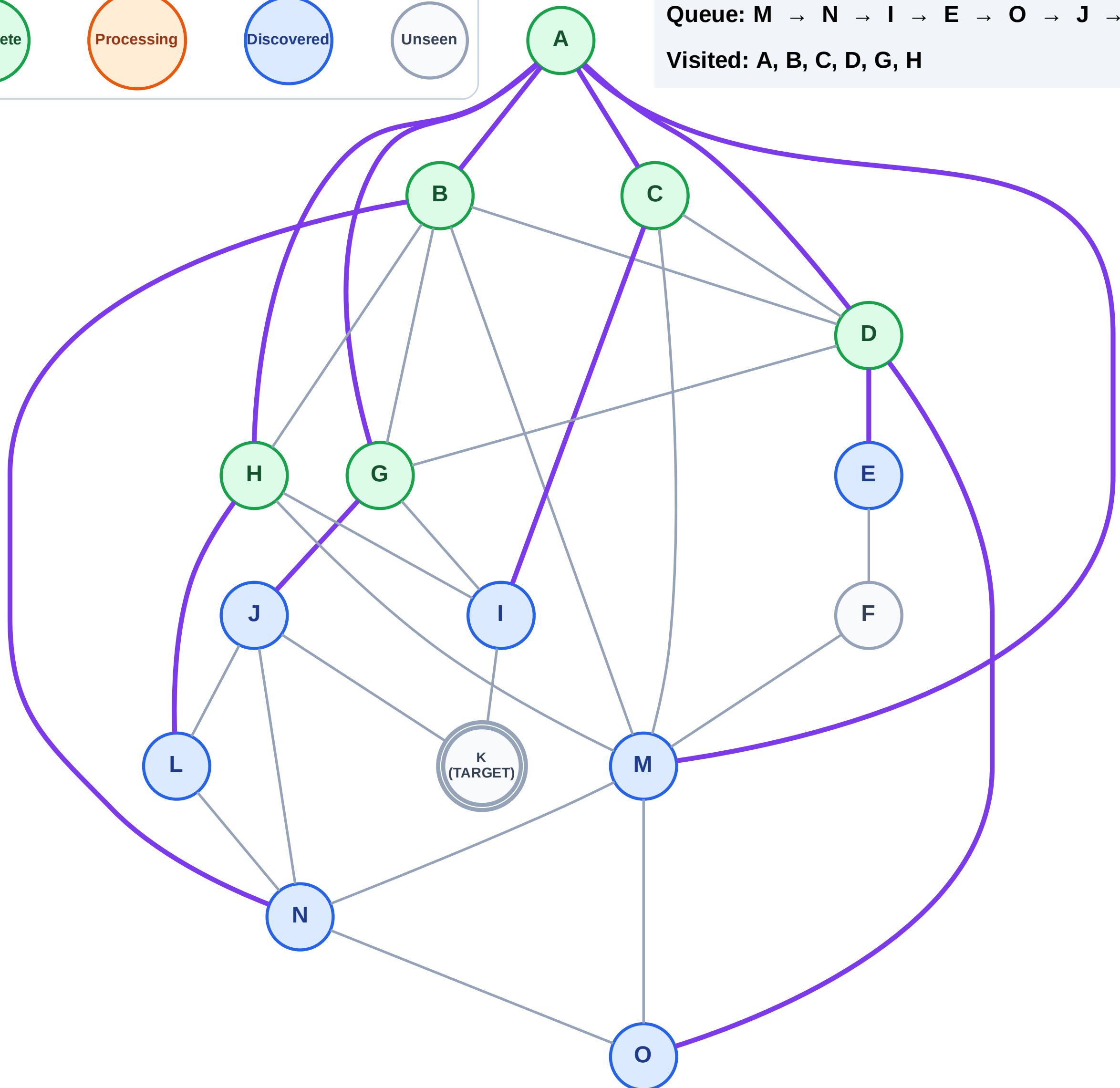
Step 13: Discover L from H

Legend



Queue: M → N → I → E → O → J → L

Visited: A, B, C, D, G, H



BFS Traversal

Step 14: Process node M

Legend

Complete

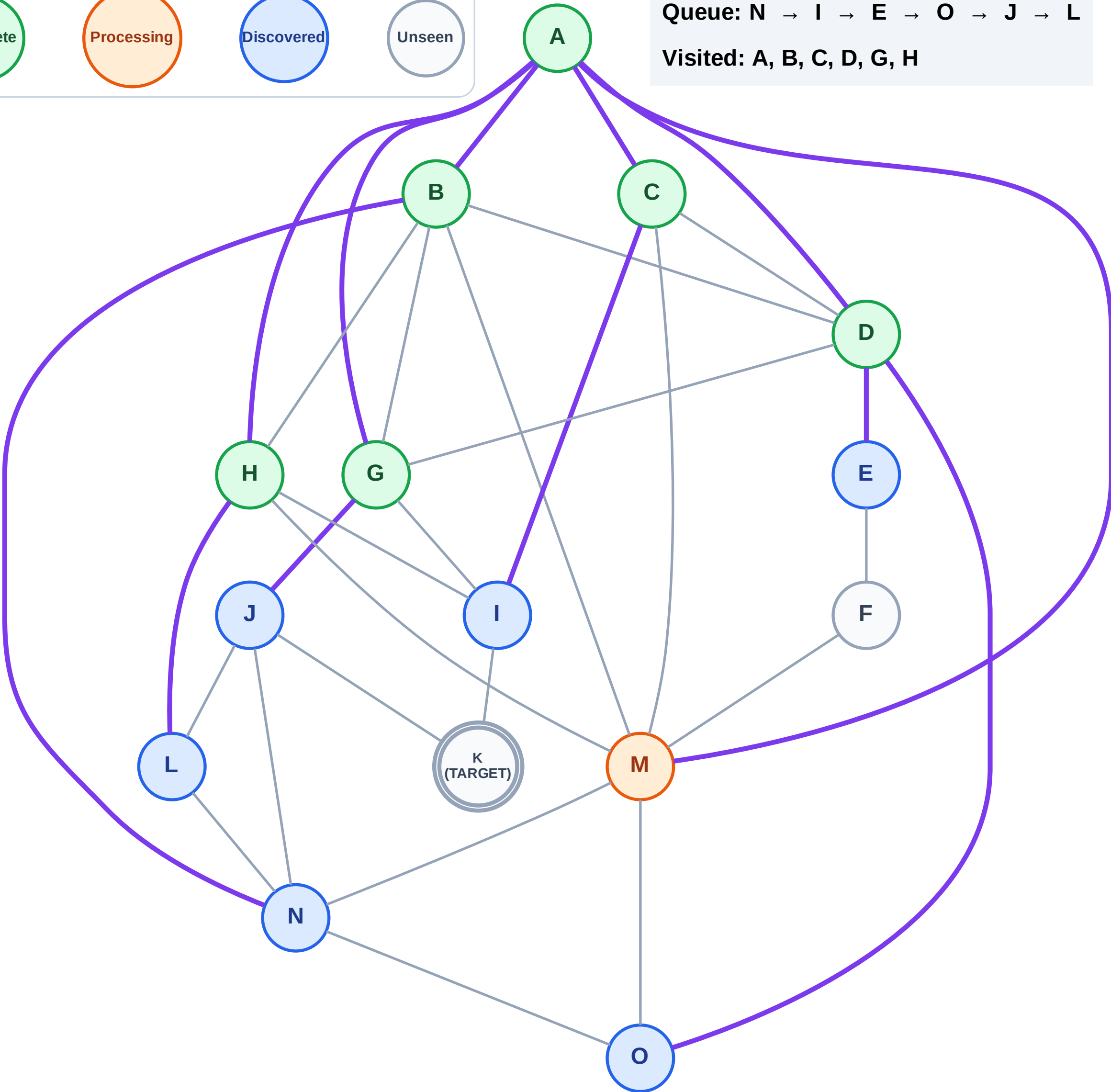
Processing

Discovered

Unseen

Queue: N → I → E → O → J → L

Visited: A, B, C, D, G, H



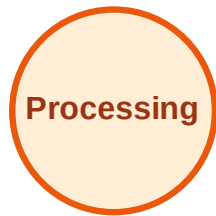
BFS Traversal

Step 15: Discover F from M

Legend



Complete



Processing



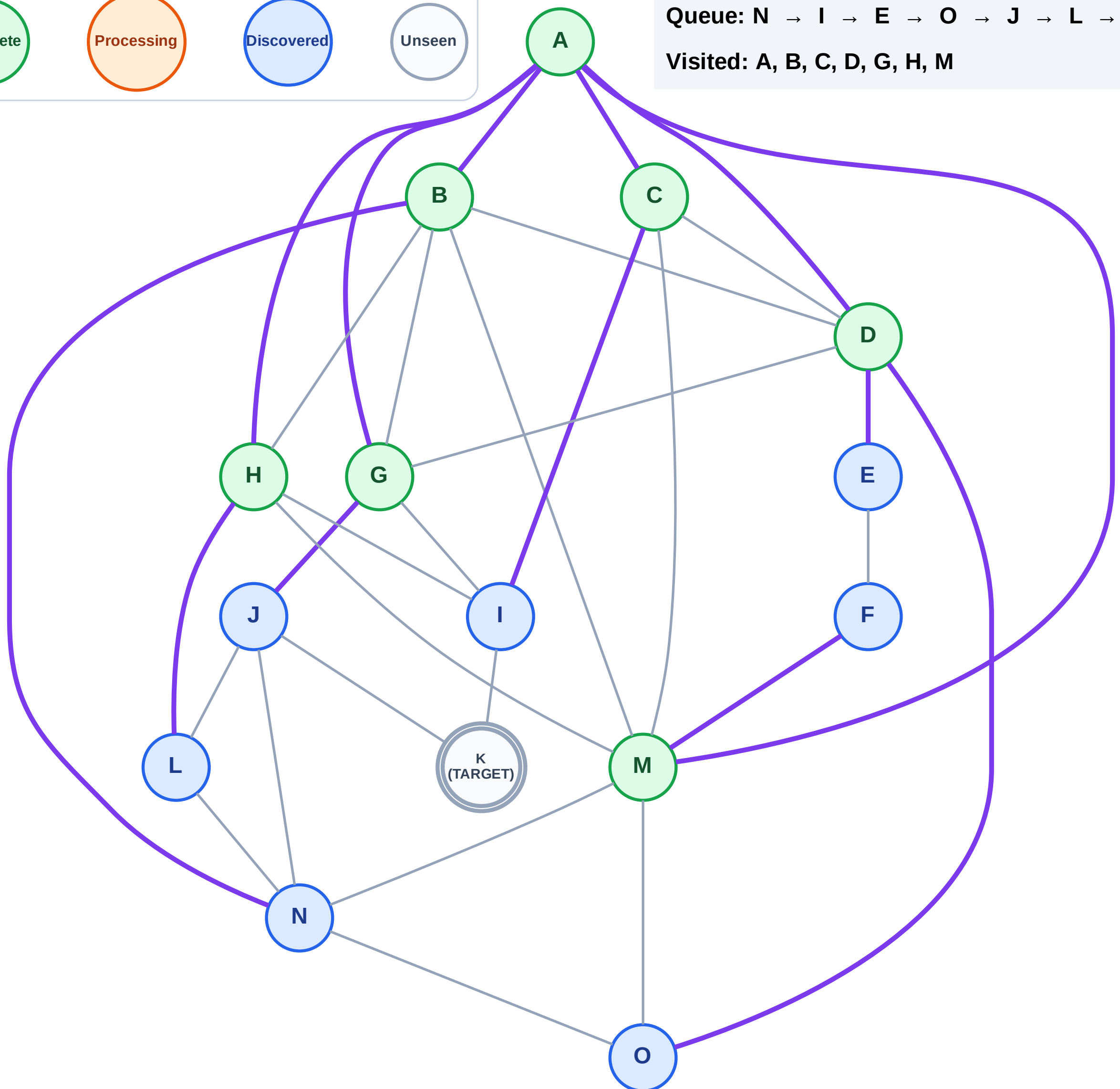
Discovered



Unseen

Queue: N → I → E → O → J → L → F

Visited: A, B, C, D, G, H, M



BFS Traversal

Step 16: Process node N

Legend

Complete

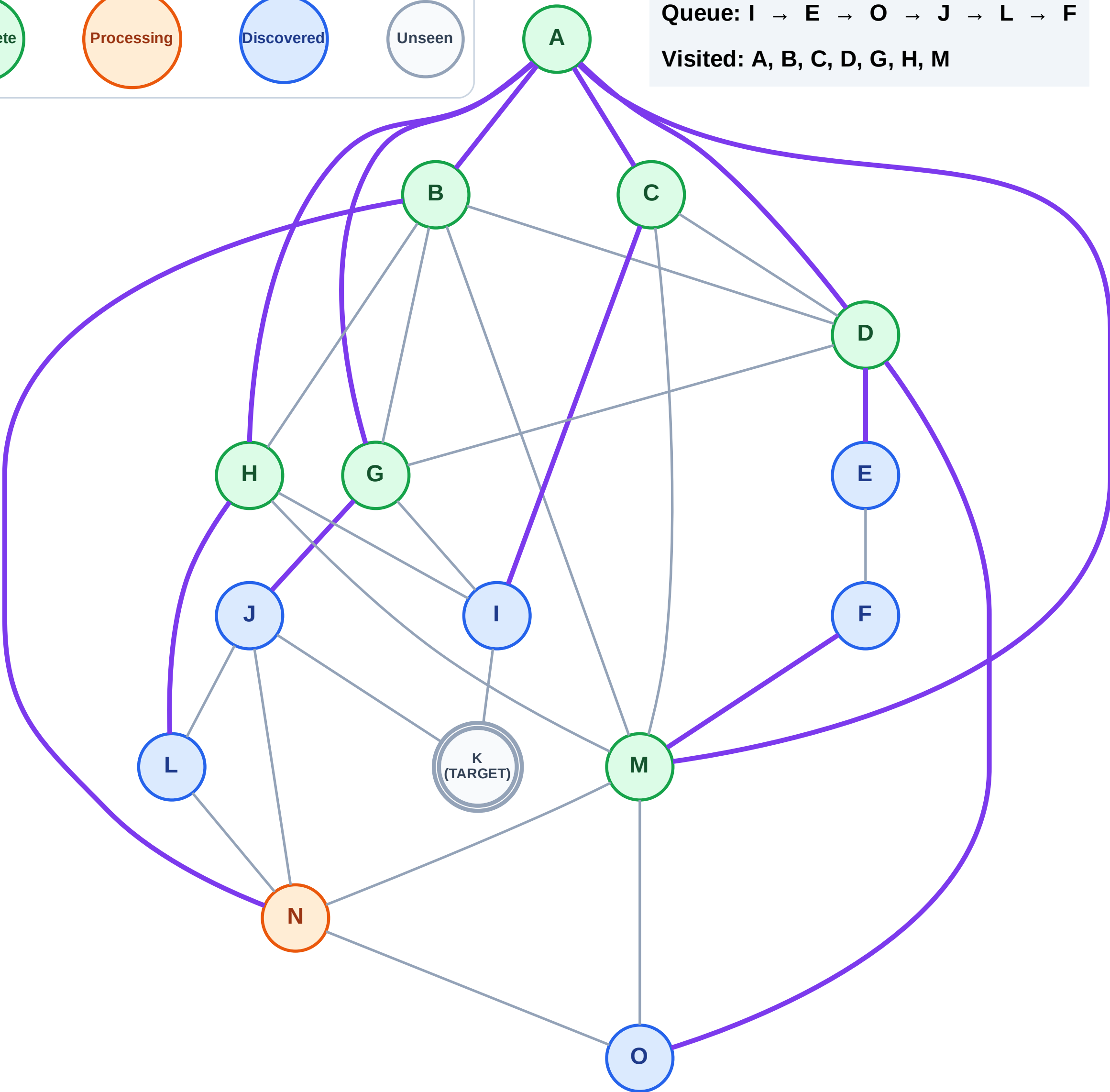
Processing

Discovered

Unseen

Queue: I → E → O → J → L → F

Visited: A, B, C, D, G, H, M



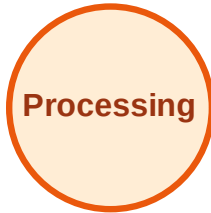
BFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



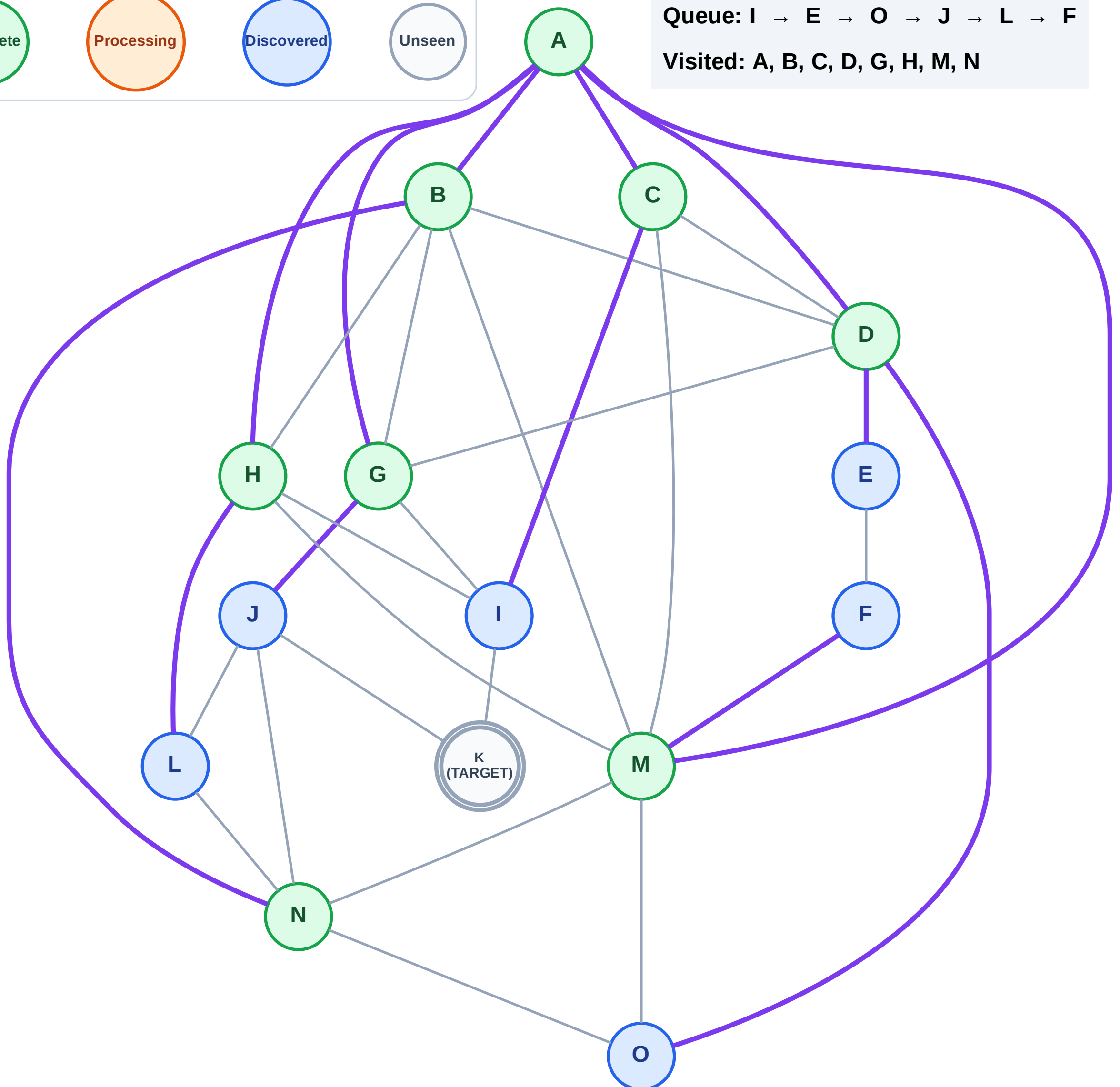
Discovered



Unseen

Queue: I → E → O → J → L → F

Visited: A, B, C, D, G, H, M, N



BFS Traversal

Step 18: Process node I

Legend

Complete

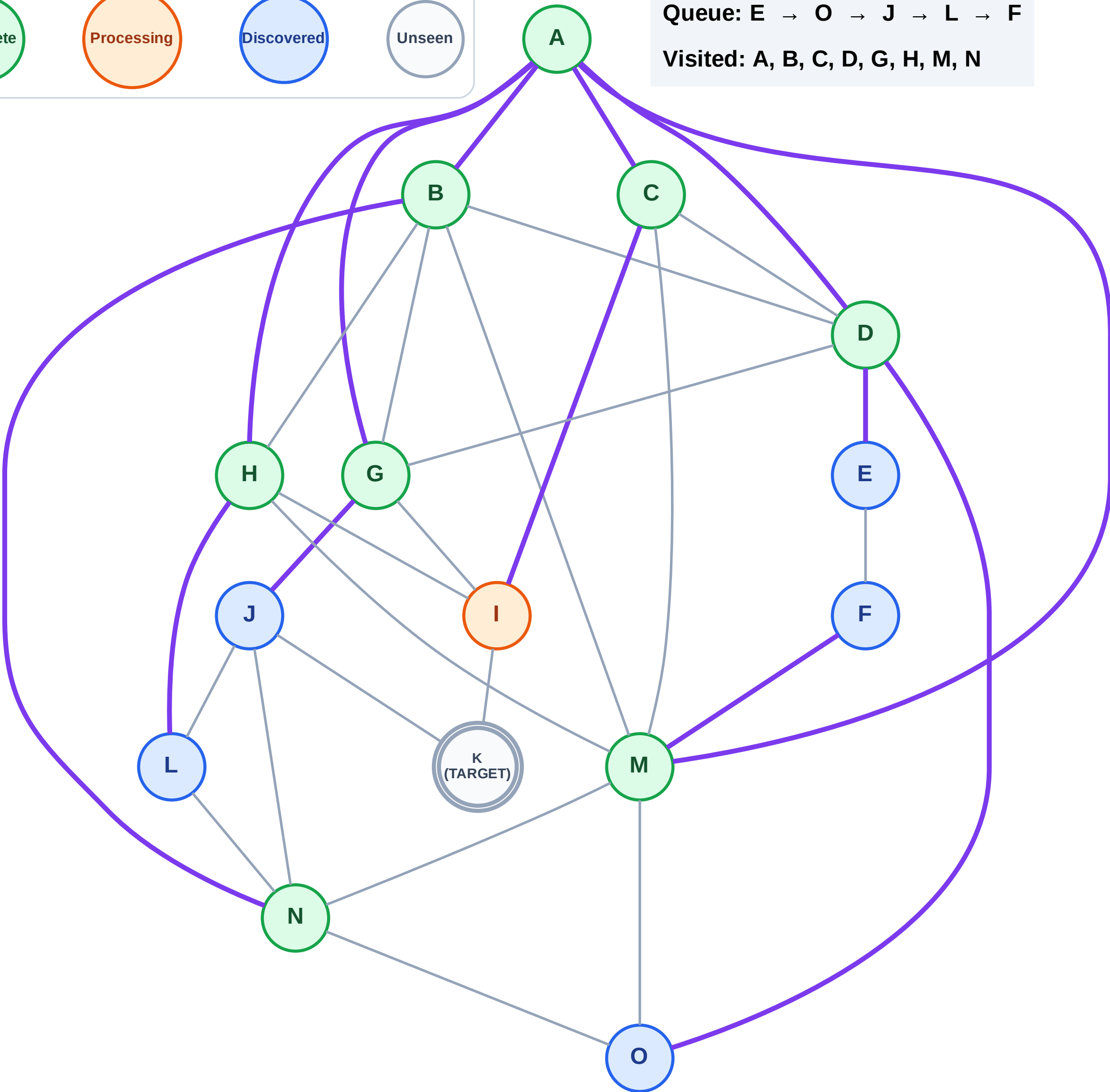
Processing

Discovered

Unseen

Queue: E → O → J → L → F

Visited: A, B, C, D, G, H, M, N



BFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → C → I → K

Legend

Complete

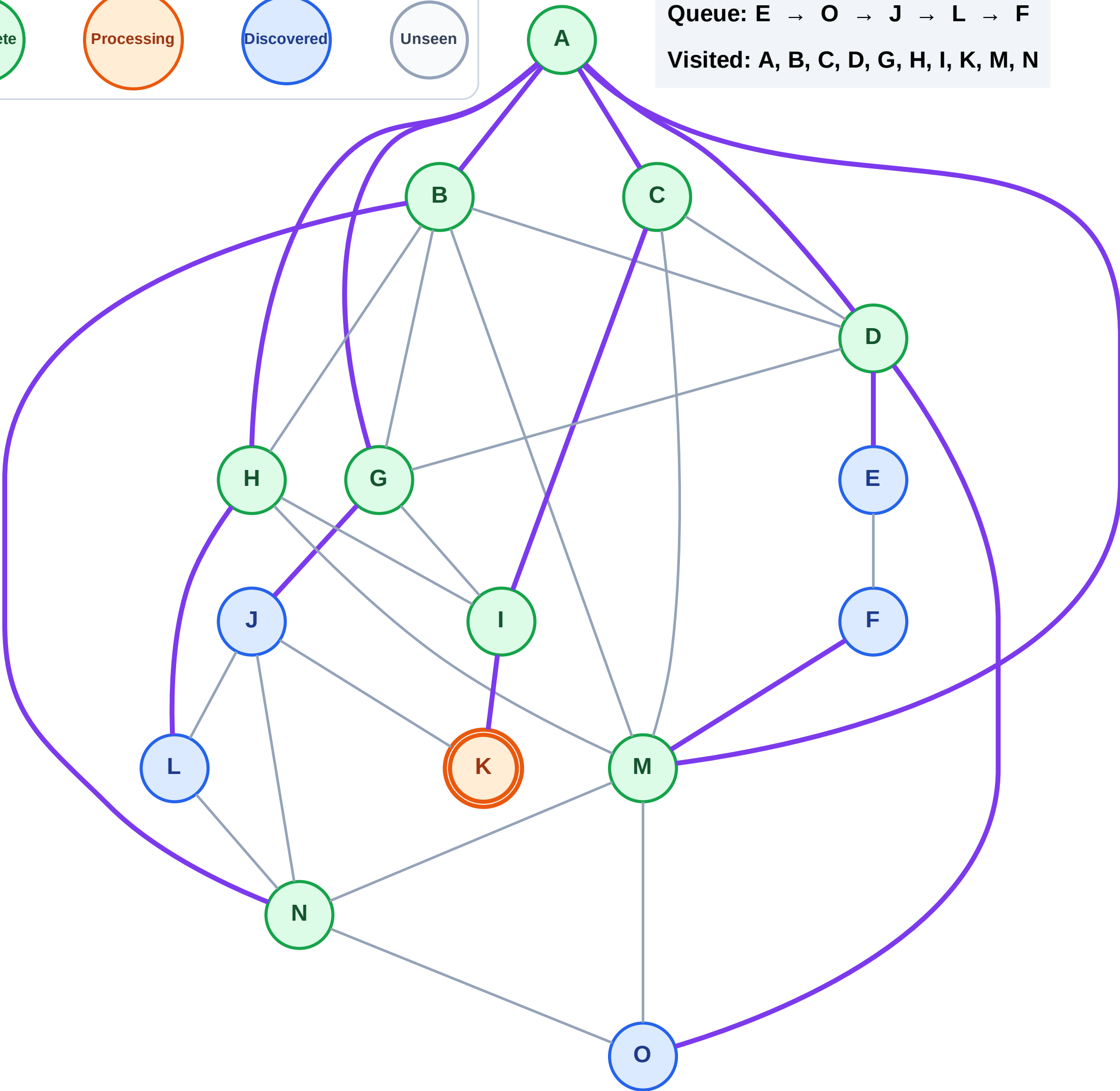
Processing

Discovered

Unseen

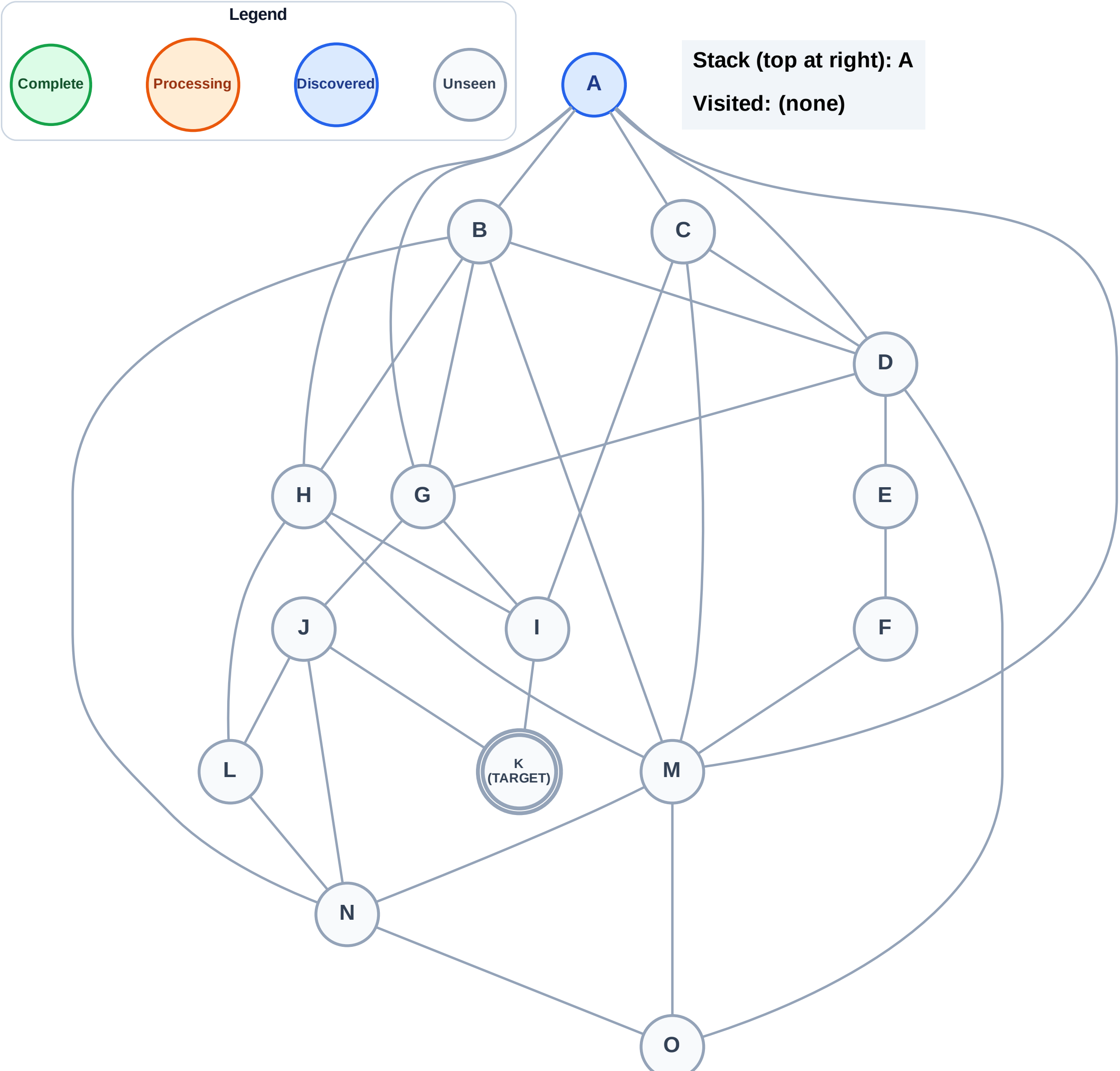
Queue: E → O → J → L → F

Visited: A, B, C, D, G, H, I, K, M, N



DFS Traversal

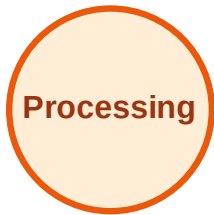
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



DFS Traversal

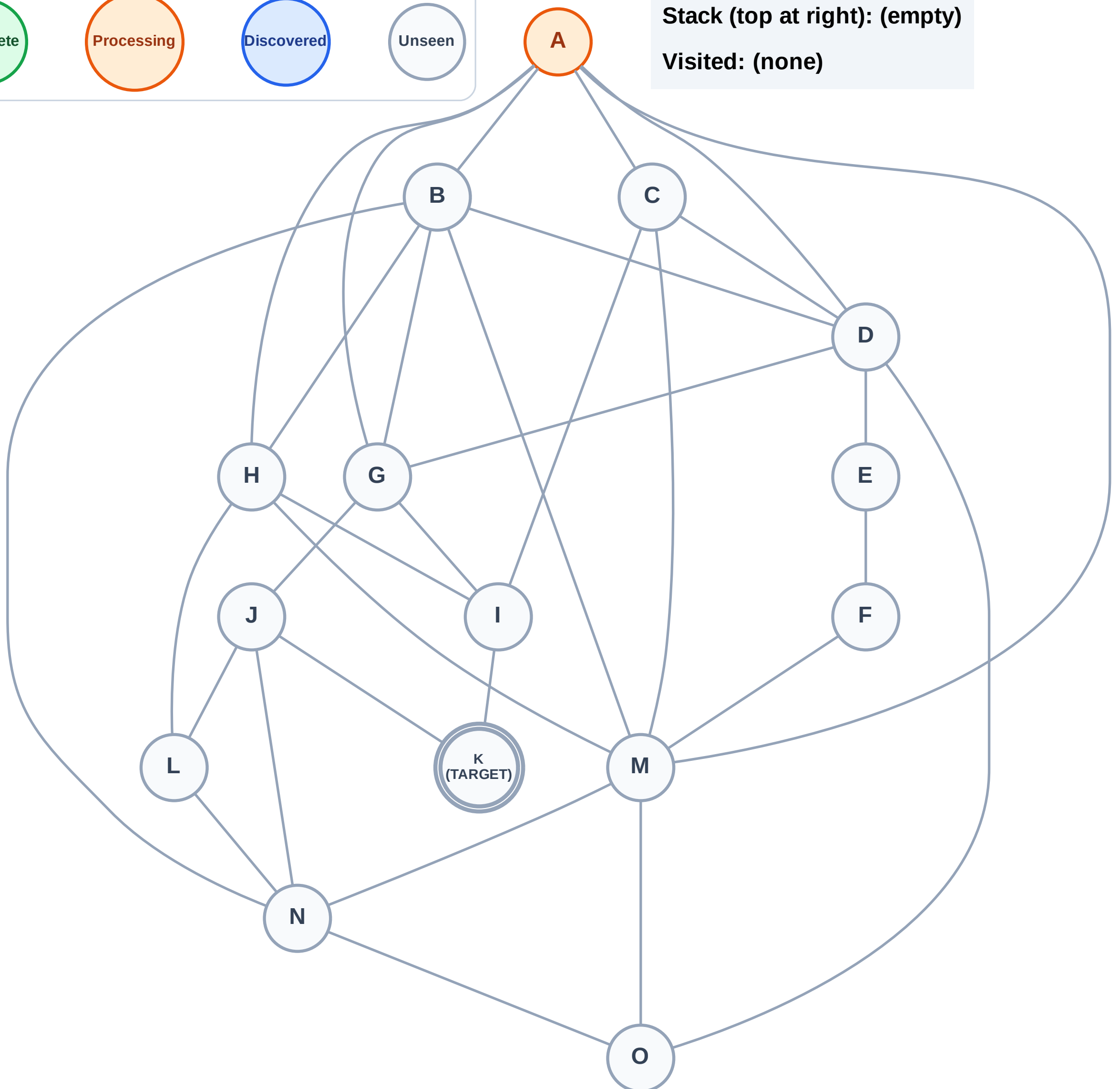
Step 2: Process node A

Legend



Stack (top at right): (empty)

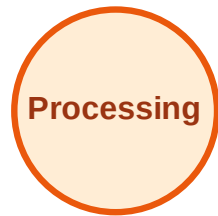
Visited: (none)



DFS Traversal

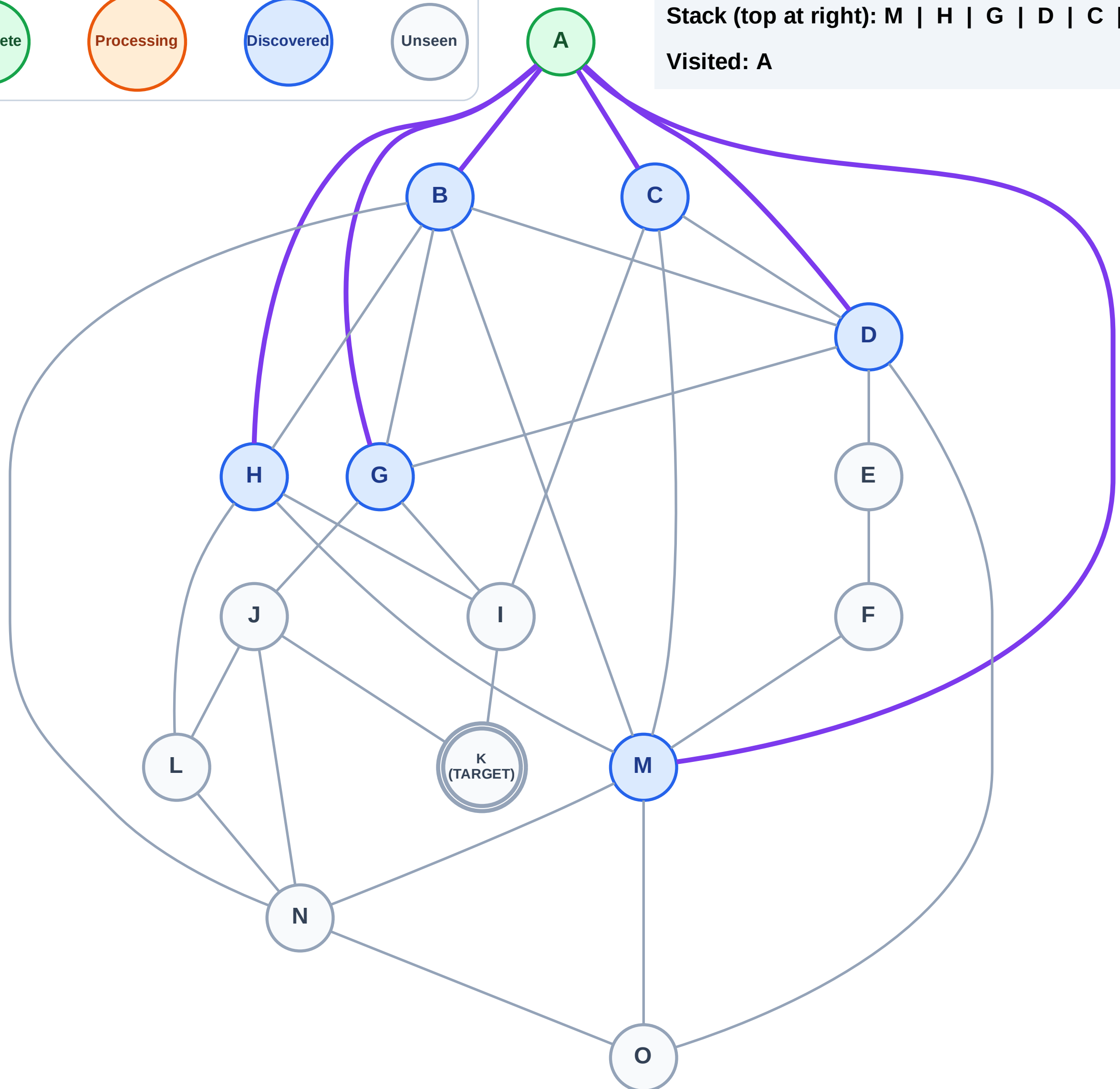
Step 3: Discover M, H, G, D, C, B from A

Legend



Stack (top at right): M | H | G | D | C | B

Visited: A



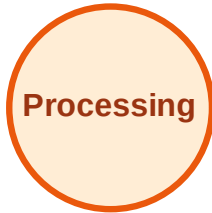
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



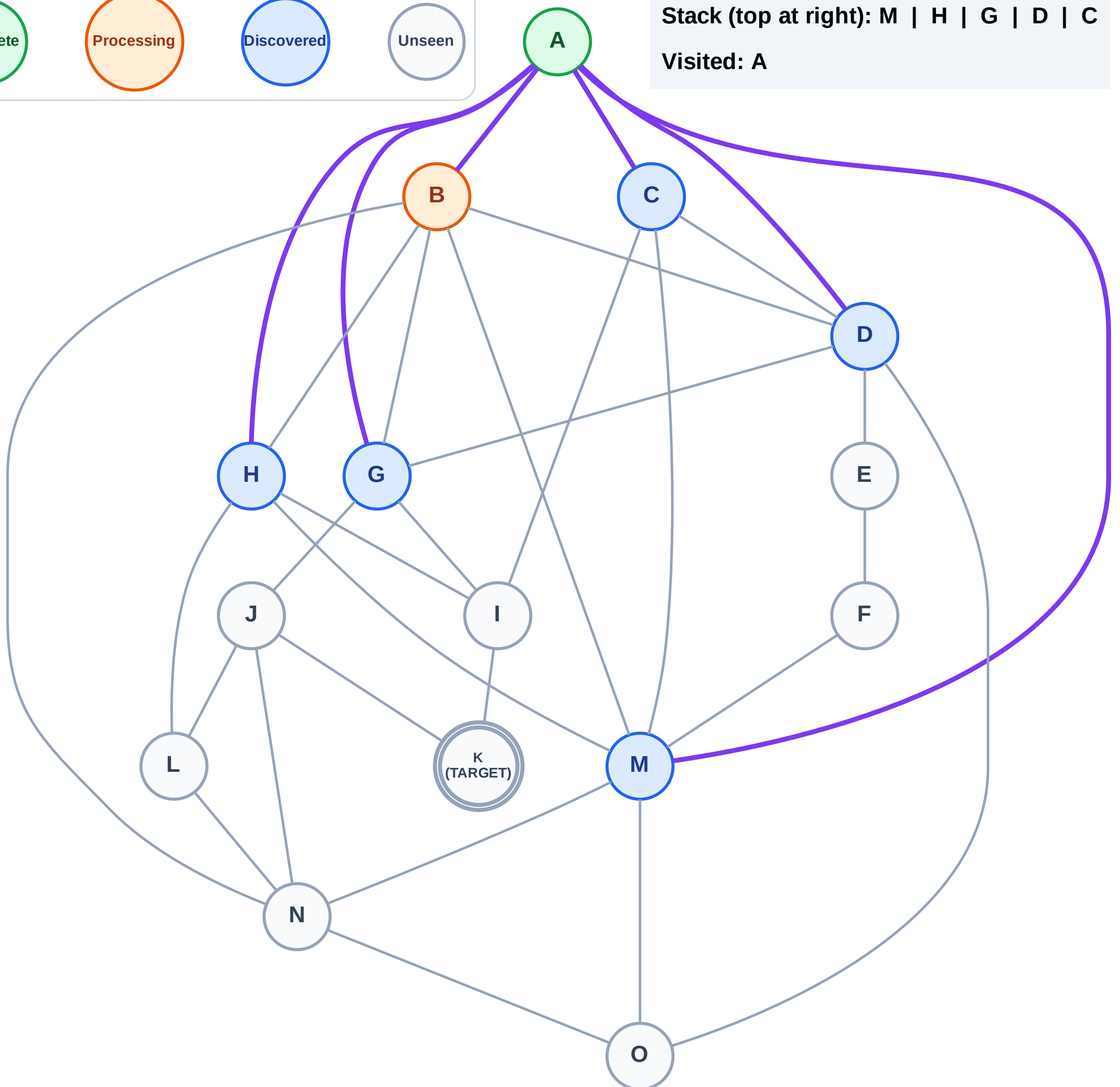
Discovered



Unseen

Stack (top at right): M | H | G | D | C

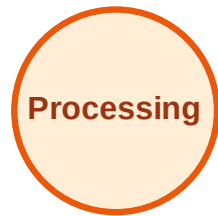
Visited: A



DFS Traversal

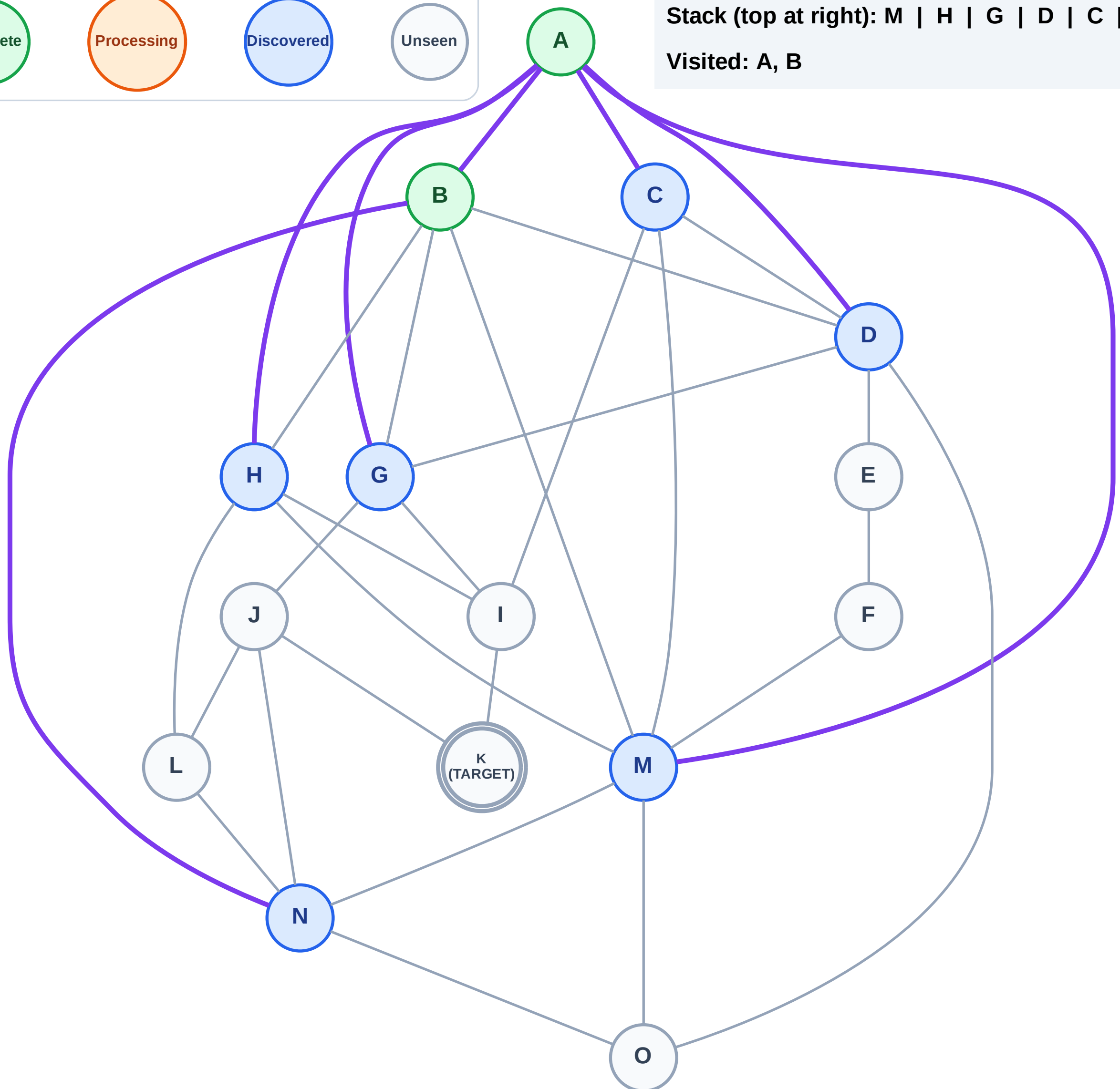
Step 5: Discover N from B

Legend



Stack (top at right): M | H | G | D | C | N

Visited: A, B



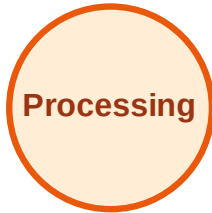
DFS Traversal

Step 6: Process node N

Legend



Complete



Processing



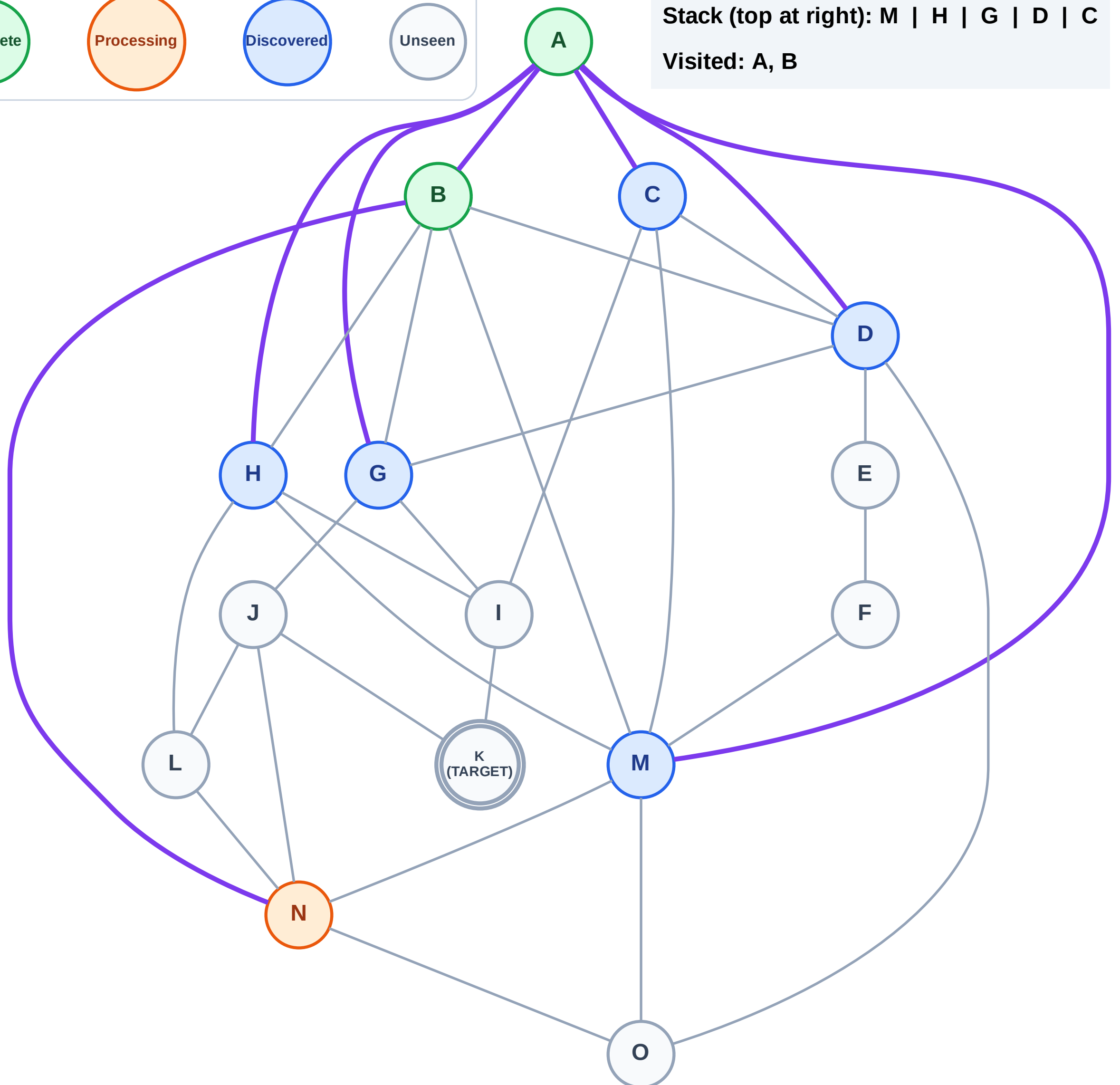
Discovered



Unseen

Stack (top at right): M | H | G | D | C

Visited: A, B



DFS Traversal

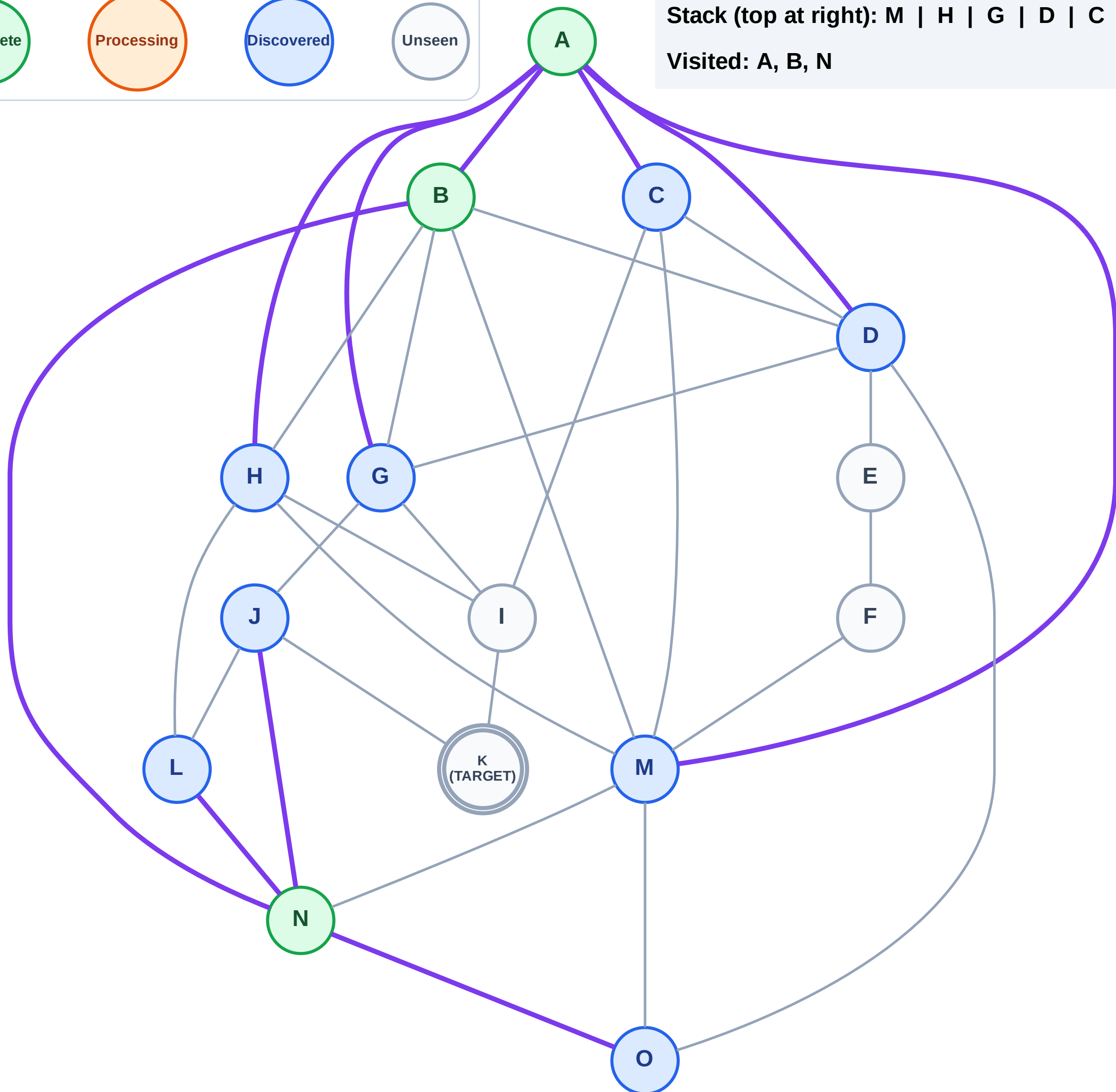
Step 7: Discover O, L, J from N

Legend



Stack (top at right): M | H | G | D | C | O | L | J

Visited: A, B, N



DFS Traversal

Step 8: Process node J

Legend



Complete



Processing



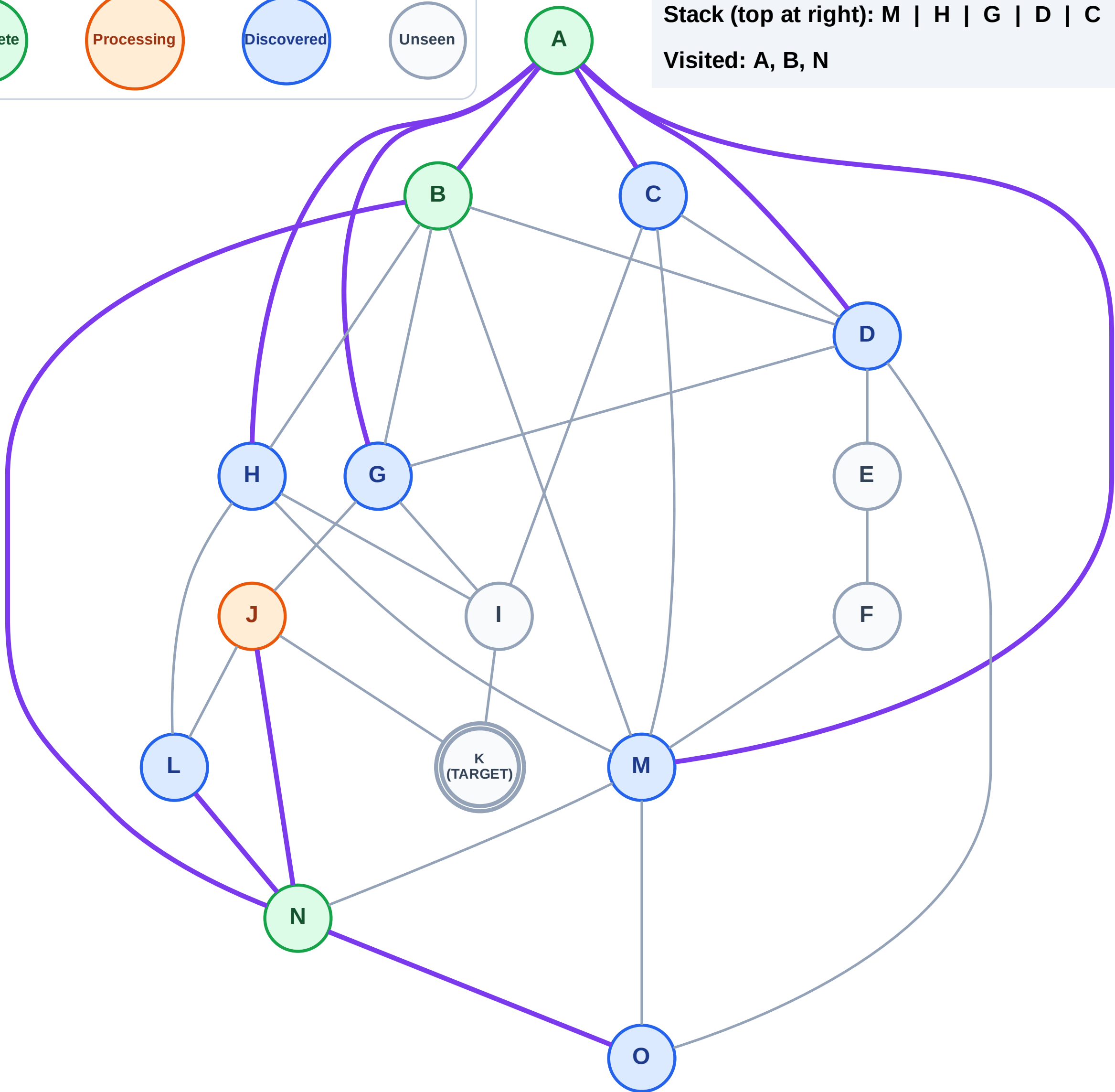
Discovered



Unseen

Stack (top at right): M | H | G | D | C | O | L

Visited: A, B, N



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → N → J → K

Legend



Stack (top at right): M | H | G | D | C | O | L

Visited: A, B, J, K, N

